# MSG No. 50.287 $P_Cban$ [ Type IV, orthorhombic ]

Table 1: Wyckoff site: 2a, site symmetry: m'm'm'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, 0\right]$ | [1,2,3,4,13,14,15,16] |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, 0\right]$ | [5,6,7,8,9,10,11,12] |

Table 2: Wyckoff site: 2b, site symmetry: m'm'm'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | [1,2,3,4,13,14,15,16] |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | [5,6,7,8,9,10,11,12] |

Table 3: Wyckoff site: 2c, site symmetry: m'm'm'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | [1,2,3,4,13,14,15,16] |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | [5,6,7,8,9,10,11,12] |

Table 4: Wyckoff site: 2d, site symmetry: m'm'm'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | [1,2,3,4,13,14,15,16] |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | [5,6,7,8,9,10,11,12] |

Table 5: Wyckoff site: 4e, site symmetry: ..2'/m'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | [1,5,12,16] |
| 2 | $\left[\frac{1}{2}, 0, 0\right]$ | [2,6,11,15] |
| 3 | $\left[0, \frac{1}{2}, 0\right]$ | [3,7,10,14] |
| 4 | $[0, 0, 0]$ | [4,8,9,13] |

Table 6: Wyckoff site: 4f, site symmetry: ..2'/m'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | [1,5,12,16] |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | [2,6,11,15] |
| 3 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | [3,7,10,14] |
| 4 | $\left[0, 0, \frac{1}{2}\right]$ | [4,8,9,13] |

Table 7: Wyckoff site: `4g`, site symmetry: `2m'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{4},\ 0\right]$ | `[1,2,15,16]` |
| 2 | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ 0\right]$ | `[3,4,13,14]` |
| 3 | $\left[-x,\ \frac{3}{4},\ 0\right]$ | `[5,6,11,12]` |
| 4 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ 0\right]$ | `[7,8,9,10]` |

Table 8: Wyckoff site: `4h`, site symmetry: `2m'm'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[1,2,15,16]` |
| 2 | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[3,4,13,14]` |
| 3 | $\left[-x,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[5,6,11,12]` |
| 4 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[7,8,9,10]` |

Table 9: Wyckoff site: `4i`, site symmetry: `m'2m'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ 0\right]$ | `[1,3,14,16]` |
| 2 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ 0\right]$ | `[2,4,13,15]` |
| 3 | $\left[\frac{3}{4},\ -y,\ 0\right]$ | `[5,7,10,12]` |
| 4 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ 0\right]$ | `[6,8,9,11]` |

Table 10: Wyckoff site: `4j`, site symmetry: `m'2m'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ \frac{1}{2}\right]$ | `[1,3,14,16]` |
| 2 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | `[2,4,13,15]` |
| 3 | $\left[\frac{3}{4},\ -y,\ \frac{1}{2}\right]$ | `[5,7,10,12]` |
| 4 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[6,8,9,11]` |

Table 11: Wyckoff site: `4k`, site symmetry: `m'm'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$ | `[1,4,14,15]` |
| 2 | $\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$ | `[2,3,13,16]` |
| 3 | $\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$ | `[5,8,10,11]` |
| 4 | $\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$ | `[6,7,9,12]` |

Table 12: Wyckoff site: `4l`, site symmetry: `m'm'2`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$ | `[1,4,14,15]` |

*continued ...*

Table 12

| No. | position | mapping |
|---|---|---|
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | [2,3,13,16] |
| 3 | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | [5,8,10,11] |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | [6,7,9,12] |

Table 13: Wyckoff site: 8m, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2}, \frac{1}{2}, z\right]$ | [1,12] |
| 2 | $\left[\frac{1}{2}, 0, -z\right]$ | [2,11] |
| 3 | $\left[0, \frac{1}{2}, -z\right]$ | [3,10] |
| 4 | $[0, 0, z]$ | [4,9] |
| 5 | $\left[\frac{1}{2}, \frac{1}{2}, -z\right]$ | [5,16] |
| 6 | $\left[\frac{1}{2}, 0, z\right]$ | [6,15] |
| 7 | $\left[0, \frac{1}{2}, z\right]$ | [7,14] |
| 8 | $[0, 0, -z]$ | [8,13] |

Table 14: Wyckoff site: 8n, site symmetry: m'..

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, y, z\right]$ | [1,14] |
| 2 | $\left[\frac{1}{4}, \frac{1}{2} - y, -z\right]$ | [2,13] |
| 3 | $\left[\frac{1}{4}, y, -z\right]$ | [3,16] |
| 4 | $\left[\frac{1}{4}, \frac{1}{2} - y, z\right]$ | [4,15] |
| 5 | $\left[\frac{3}{4}, -y, -z\right]$ | [5,10] |
| 6 | $\left[\frac{3}{4}, y + \frac{1}{2}, z\right]$ | [6,9] |
| 7 | $\left[\frac{3}{4}, -y, z\right]$ | [7,12] |
| 8 | $\left[\frac{3}{4}, y + \frac{1}{2}, -z\right]$ | [8,11] |

Table 15: Wyckoff site: 8o, site symmetry: .m'.

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, \frac{1}{4}, z\right]$ | [1,15] |
| 2 | $\left[x, \frac{1}{4}, -z\right]$ | [2,16] |
| 3 | $\left[\frac{1}{2} - x, \frac{1}{4}, -z\right]$ | [3,13] |
| 4 | $\left[\frac{1}{2} - x, \frac{1}{4}, z\right]$ | [4,14] |
| 5 | $\left[-x, \frac{3}{4}, -z\right]$ | [5,11] |
| 6 | $\left[-x, \frac{3}{4}, z\right]$ | [6,12] |
| 7 | $\left[x + \frac{1}{2}, \frac{3}{4}, z\right]$ | [7,9] |
| 8 | $\left[x + \frac{1}{2}, \frac{3}{4}, -z\right]$ | [8,10] |

Table 16: Wyckoff site: `8p`, site symmetry: `..m'`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ 0]$ | `[1,16]` |
| 2 | $\left[x,\ \frac{1}{2}-y,\ 0\right]$ | `[2,15]` |
| 3 | $\left[\frac{1}{2}-x,\ y,\ 0\right]$ | `[3,14]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$ | `[4,13]` |
| 5 | $[-x,\ -y,\ 0]$ | `[5,12]` |
| 6 | $\left[-x,\ y+\frac{1}{2},\ 0\right]$ | `[6,11]` |
| 7 | $\left[x+\frac{1}{2},\ -y,\ 0\right]$ | `[7,10]` |
| 8 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$ | `[8,9]` |

Table 17: Wyckoff site: `8q`, site symmetry: `..m'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ y,\ \frac{1}{2}\right]$ | `[1,16]` |
| 2 | $\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | `[2,15]` |
| 3 | $\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}\right]$ | `[3,14]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$ | `[4,13]` |
| 5 | $\left[-x,\ -y,\ \frac{1}{2}\right]$ | `[5,12]` |
| 6 | $\left[-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[6,11]` |
| 7 | $\left[x+\frac{1}{2},\ -y,\ \frac{1}{2}\right]$ | `[7,10]` |
| 8 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$ | `[8,9]` |

Table 18: Wyckoff site: `16r`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $\left[x,\ \frac{1}{2}-y,\ -z\right]$ | `[2]` |
| 3 | $\left[\frac{1}{2}-x,\ y,\ -z\right]$ | `[3]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$ | `[4]` |
| 5 | $[-x,\ -y,\ -z]$ | `[5]` |
| 6 | $\left[-x,\ y+\frac{1}{2},\ z\right]$ | `[6]` |
| 7 | $\left[x+\frac{1}{2},\ -y,\ z\right]$ | `[7]` |
| 8 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$ | `[8]` |
| 9 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z\right]$ | `[9]` |
| 10 | $\left[x+\frac{1}{2},\ -y,\ -z\right]$ | `[10]` |
| 11 | $\left[-x,\ y+\frac{1}{2},\ -z\right]$ | `[11]` |
| 12 | $[-x,\ -y,\ z]$ | `[12]` |
| 13 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ -z\right]$ | `[13]` |
| 14 | $\left[\frac{1}{2}-x,\ y,\ z\right]$ | `[14]` |
| 15 | $\left[x,\ \frac{1}{2}-y,\ z\right]$ | `[15]` |
| 16 | $[x,\ y,\ -z]$ | `[16]` |